

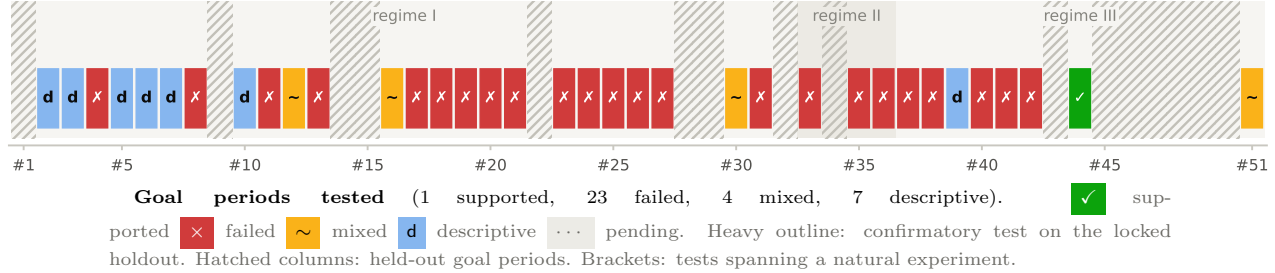
H55: Norm-enforcers are the swarm's immune cells

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

Bottom line. Correctors get warm replies, not resistance; corrections almost never reach looping agents (no immune response); plain attention is the only loop-breaker found.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

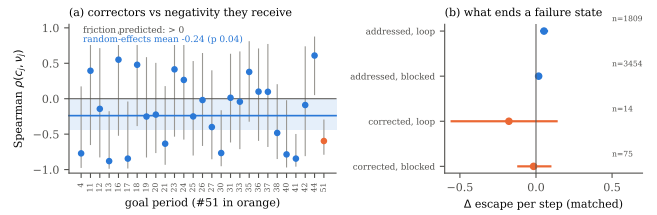
The test asks whether agents who correct others draw negative stance, and whether corrections end self-repetition loops and blocked spells. Corrections come from DQ2's confident Jev correction/decline subtype. A blind check gives precision 0.93 and recall about 0.1; the H55 lexical marker fails (0.30). Over 27 periods, correctors receive *warmer* replies (random-effects $\rho = -0.24$, $p = 0.04$; #51 -0.60). Replies to a correction are no colder than replies to the same agent's other messages. Corrections reach a looping agent in 0.65% of loop episodes. Matched contrasts show no effect on escape (loops -0.18 , blocked -0.02 ; underpowered). Being addressed at all raises loop escape by $+0.05$. #51's assigned enforcers do not correct more. Nobody enforces #16's operator rules. **Round 1c** uses stance v2.1: corrections are the **correct/decline** classes (recall about 0.7, precision about 0.5) and friction is the validated disagreement flag, with the labeller's confusion in every null. Correctors receive slightly *less* disagreement ($\rho = -0.16$, $p = 0.047$). With $4\times$ the treated events, a read correction still does not end blocked spells ($\Delta = +0.007 [-0.053, +0.070]$, power 0.96 at 1 logit). The verdict is **failed**. The holdout is not run.

PHYSICAL MODEL

Escape from a failure state (restatement loop or v3 "blocked" spell) is a discrete-time hazard with frailty α_i and loop age k . Model 06 gives the neutral null. D marks a message addressed to the agent since its last step; C marks a correction among them. Immunity means $\beta_C > 0$ beyond β_D . Friction means the received-stance field $\nu_j = -b_j$ (ordered logit, replier fields a_i) rises with the correction rate c_j . The immune contrast measures the viability value ΔV of corrections (model 04).

$$\text{logit } h_i(k) = \alpha_i + \beta_{\text{age}} \ln k + \beta_D D + \beta_C C$$

$$P(y_{ij} \leq m) = \sigma(c_m - a_i - b_j), \quad \nu_j = -b_j, \quad \rho(c_j, \nu_j) > 0$$



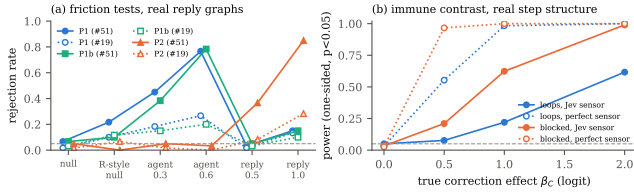
Observables. (a) Per-period Spearman ρ between an agent's correction rate and the negativity it receives (Fisher- z 95% intervals; band: random-effects mean). Friction predicts $\rho > 0$; the swarm shows the reverse. (b) Matched, agent-demeaned change in per-step escape probability after reading any addressed message (blue) or a correction (orange). Attention ends loops a little. Corrections are too rare (0.65% of loops) to show an effect.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To break a loop, address the looping agent by name. Each addressed message raises its per-step escape probability by 5 points, and more in debates. Do not expect peers to correct a stuck agent. One loop in 150 receives a correction, and peer enforcement of an operator rule stays at background. Supply error correction from outside the peer network: the operator, the nudger or a designated reviewer. Peer correction costs no social standing: correctors receive warmer replies, so enforcers need no protection from ostracism. HH210's "missing immune system" holds in the coverage sense.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 correctors negativity	draw meta $\rho = -0.24$ $[-0.44, -0.02]$, 10/27 > 0	reversed
P1b beyond speaker field	own -0.24 (same)	fail
P2 replies to corrections colder	$\gamma = -0.006$ $[-0.09, +0.08]$	fail
P3 received corrections are corrections	opposes median 0.66 (21/27 ≥ 0.5)	pass
P4 corrections loops	end $\Delta = -0.18$ $[-0.56, +0.14]$, 14 matched	underpowered
P5 corrections blocked spells	end $\Delta = -0.02$ $[-0.12, +0.10]$, 75 matched	fail (no large effect)
P6 any addressed ends loops	read $+0.053$ $[+0.028, +0.080]$, $p = 10^{-4}$	pass
P7 more enforcement, shorter loops	$\rho = -0.24$, $p = 0.24$ (not identifiable)	fail
P8 $< 20\%$ of loops corrected	0.65% (35% addressed)	pass
Native G51/G12/G16/G38	roles not enacted; cooler; no peer enforcement; no address effect in #38	judges mixed $\times 3$, failed
1c P1 (v2.1)	$\rho = -0.16$, $p = 0.047$; meta reversed power 0.83	
1c P2 (v2.1 flag)	$\gamma^D = +0.27$ pp (SE 0.16), NG $p = 0.051$	not passed
1c P4 loops	$+0.05$ $[-0.12, +0.21]$, treated	93 underpowered
1c P5 blocked	$+0.007$ $[-0.053, +0.070]$, treated	fail (powered)
1c P8 coverage	2.9% of loops (true $\approx 2\%$)	pass



Second figure. Synthetic validation on real reply graphs and real loop and blocked step structures, with the measured sensor and stance noise. (a) Friction tests: calibrated under the null. P1 also fires under reciprocity (R-style), which the partial P1b controls. (b) Power of the immune contrast: the sensor’s recall limits it, not the design (dotted: perfect sensor).

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 $\times$ failed
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 $\times$ failed
G10 d descriptive
G11 $\times$ failed
G12 $\sim$ mixed
+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Round 1’s correction sensor saw one correction in ten. The v2.1 sensor sees about seven in ten, but half its flags are overcalls, so c_j mixes correcting with status reporting. One blind rater did the validation, with a broader rubric than DQ2’s. Stance is reliable only in aggregate. The warmth toward correctors can partly be a labeller habit. Being addressed may end a restatement mechanically, because an answer is not a restatement. Loops are within-day restatement runs, many of them benign. #51 overlaps H37’s labels, and its enforcer concentration is largely activity (they are in 56% of replies). The synthetic check finds and removes a targeting confound with agent fixed effects; within-agent targeting remains.

ROUND 2

Raise the precision of the correction class (a confident-correct rule validated blind) to power the loop test. Test whether correctors are right more often (work ledger). Re-freeze `confirm.py` on the v2.1 held-out labels after Vivian’s sign-off.

- **Essence.** whether an agent swarm repairs its own errors, and what an operator must supply when it does not.
- **H55-R1** A high-recall correction sensor.
- **H55-R2** Deference, not friction.
- **H55-R3** Attention is the lever.
- **H55-R4** HH210 quantitatively.